Accessibility of Community Flyers: A Content Analysis of Social Service Flyers in Utah County

Grace Smith, Julia Jensen, and Sage Uchytil

Linguistics Department, Brigham Young University

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Dr. Grant Eckstein

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Introduction

Community-based social service organizations are at the front lines of providing urgent relief to low-income residents in the United States. In Utah, the greater Provo area has over 230 human-centered non-profit organizations. This number comprises community health, food assistance, literacy and education, family and children services, mental health, resource navigation, and legal support service organizations (Cause IQ, n.d.). By nature of the services one can logically infer that the services provided are intended for low-income individuals and families who do not have readily available resources provided by employers or extended family, especially immigrant and refugee newcomers to the United States. Indeed, based on the organization's descriptive information, of those 232 non-profit organizations, 9-10 directly service immigrants and refugees, and 23-46 more broadly provide services from which this population benefits. That is between 10%-20% of the non-profit's targeted demographic of support (Refugee Services Office, n.d.). The Federal Administration for Children & Families, Office of Refugee Resettlement's 2023 annual survey found 50% of refugees reported that they had no prior knowledge or experience with English at all, print or speaking, and 35% reported their language proficiency as "not well" (Administration for Children and Families, 2024). Additionally, according to the National Center for Education Statistics and the Programme for the International Assessment of Adult Competencies (PIAAC) survey, roughly 17% of Salt Lake County adults are at or below a level 1 in literacy. Comparatively, Utah county has roughly 12% of adults at or below a level 1 literacy (National Center for Education Statistics [NCES], n.d.). The PIAAC uses its own statistical estimate models to arrive at level descriptions for English proficiency but the lens through which we are assessing the flyers relevant to the current study are the federal NRS levels and the internationally widely-used CEFR levels of English

proficiency. In a comparison of these three models, a PIAAC level 1 or lower is comparable to an NRS beginning literacy/low beginning ABE, and high beginning level. A PIAAC level 1 or lower is comparable to a CEFR pre-A1 to A2. These comparisons describe a low-proficiency, English language learner as someone who can read very short, simple pieces of information. This type of community member needs clear, easily-identifiable markers of language and graphics to understand a general meaning of a text.

Research Question and Literature Review

As English language instructors, our question is how linguistically and visually accessible are community-based information flyers to low-level English learners in the greater Provo area? We hope the collected data, and subsequent analysis and discussion can inform local community-based, human-service organizations in designing flyers that are readily and easily accessible to all levels of English language proficiency.

As a means to inform our analysis of the thirty collected community-based service flyers, we conducted a brief literature review of salient print and visual literacy studies on low-literacy learners to complement our own experience interacting with students and flyers. A staple in adult education visual literacy is Dan Bruski's (2012) research article, "Graphic device interpretation by low-literate adult ELLs: Do they get the picture?" He utilized philosopher Charles S. Peirce's construct of visual literacy "signs," which was based on Sandra Moriarty's work in visual communication. This construct categorizes images, or graphic devices, as iconic, indexical, and symbolic in meaning and interpretation. Although Bruski's research focused on images used in educational materials, the findings of the study have implications across adult literacy and print materials used in educating and informing adults in the community. Bruski's first point of discussion relevant to our study is that visual literacy and the ability to read and write are

different types of literacy that are not fully appreciated by the broader community. He argues that just as children learn to “read” pictures before they begin the literary process of identifying letters and sounds, adults learning to read for the first time benefit from this same process. But what if they come from a country and culture that transmits stories and information mostly in oral form? Bruski found that students’ understanding of the “graphic devices” in the educational materials came mainly from personal exposure to print, rather than from their broader cultural experience in childhood. Many learned to interpret the images because of prior experience in the English classroom in the U.S. Bruski tiered Peirce’s three types of graphic devices according to the student's ability to grasp their meaning in accordance with the results of his study (Pierce, based on Moriarty, 2004). 1) Iconic signs are images, or graphic devices, that look like the things they represent. This type of image was the easiest for the students in his study to understand. 2) Indexical signs are images that indicate the existence of something. 3) Symbolic images stand for something and are attuned to the general cultural awareness of the country or community . These images were the hardest for students to interpret due to their lack of cultural competency and experience with formal print materials. Interestingly, of the nine Somali students who participated in the study, five were not literate in their L1 and yet, showed greater competency in interpreting the iconic images than their L1 literate counterparts. Additionally, the strengths reversed when it came to symbolic images. This implies that a non-literate adult’s need to navigate their community with visual intelligence can be carried over to another culture and community but with caveats. Specific cultural “codes” fix meaning to images that not all new-comers to the community will understand (Bruski, 2012). Bruski’s research was limited in that it focused on educational materials only (Step Up curriculum series).

South African researcher, Adelia Carstens, focused her 2004 research on print materials designed for the general public. She intended to guide community and government organizations in designing materials that account for varying levels of adult literacy. Her article, "Tailoring print materials to match literacy levels: A challenge for document designers and practitioners in adult literacy," outlines the need for "plain language" in public print materials. Sandra detailed the significant low-literacy numbers among the population at the time, and the barriers this presented to communicating to the public via public health campaigns, leaflets, brochures. Carstens' research differed from Bruski's in its use of eye-tracking and semiotics literature to triangulate the data on where the eye focuses when viewing a public information document, and how linguistically accessible the language on the document was to the reader. The findings of her research showed that many flyers included significant vocabulary barriers such as medical terminology, long sentences, weak visual queues, unrealistic black and white images, depth perception ambiguity, and generally too much information. Readers experienced cognitive overload, disengagement, and confusion over artistic style input that had no meaning (Carstens, 2004). Bruski's and Carstens' research compliment each other in their findings that low-literate readers need simple, plain language, and a stylistic focus, in order to provide visual queues that provide context for the purpose of the message.

Lastly, we briefly examined the United States' Center for Disease Control's "plain language materials and resources" website. We were pleased to learn that the government passed and implemented the 2010 "Plain Writing Act." This act required federal agencies to train staff to use plain language when they communicated with the public (Center for Disease Control and Prevention, 2026). Consequently, the website included plain language resources as it relates to public health communications, NIH communication, and lists of everyday words to replace

health vernacular. It included a robust product development tool and PDF intended to support those with low-literacy and cognitive barriers, and an example of a health information poster informing the public of the Dengue vaccine. A thoughtful and practical plain language checklist is provided for community and health organizations to systematically review their public communications, including flyers to assess for visual and linguistic accessibility for all levels of the adult population.

Bruski's and Carstens' studies, in tandem with the U.S. government legislation, provided a rich and targeted perspective through which we analyzed the thirty flyers collected. Our methodology and findings are discussed below.

Methodology

The methodology used for this research is centered on the traditional procedure of content analysis. This method is appropriate for our research due to the nature of data collection from sources already present in the world. No data was created for this project, but instead, to achieve this, data was collected from thirty different flyers around the Provo/Orem area which were aimed toward English learners and immigrant communities in Utah Valley. There were no direct participants in the study, as only already existing content was analyzed, which was not necessarily tied to any individual. The flyers ranged in topics, from health clinics to workshops to invitations to community events. As the goal of this research was to determine how accessible these flyers are to pre-literate and A1 level English learners, each flyer was analyzed thoroughly and examined from a few different angles. We created an analysis code that examined total word count, average font size of main text, the font size of the titles and important information, average syllabic count per word, number of images, symbols, QR codes or links present, and

sentence length. Each of these factors provides important information about how accessible a certain flyer might be to the community of emerging English language learners.

First, with the help of Microsoft's Copilot AI system,we analyzed each of the flyers for sentence length, syllable count, total word count, and font size. We then revisited the data to confirm accuracy of information and include researcher analysis. We designed these codes in order to quantify our data. For transparency, we tasked Copilot with only counting the syntactic data points (i.e., word count, syllables, sentence length, and font size). We followed this common method of AI use in research so that we the researchers could analyze the images, graphic design, and formatting to bring human nuance to the analysis. However, all of the findings from the code were looked over thoroughly by all three researchers. We altered some of the categorization done by AI to preserve validity and accuracy because it can fail to categorize the data according to our experience and literature review. Then we categorized the agreed upon data into Table 2.

After looking at the words, syllables, and sentence length, we counted data regarding images, symbols, and QR codes on each individual flyer. This data was organized and added into Table 2 as well. These were counted by each researcher. With word and syntactic information and visual data, it was simpler to determine the accessibility level of each flyer to the target community. There was a wide scope of results found in the data, total word counts ranging from only 12 total words on a flyer to 420, and while some flyers provided many invitations to action, emails, phone numbers, links, and QR codes to know more, others did not provide any at all.

After data organization, we determined the accessibility of community flyers for emergent readers using the PIAAC levels. We categorized the rating options as follows: Very Low, Low, Moderate-Low, Moderate, Moderate-High, High, Very High. For each flyer rating, we

listed primary strengths and primary barriers of intelligibility and accessibility of the flyers. These strengths and barriers explain the reasoning for why a specific flyer is considered highly accessible or not to the emergent reader.

Table 1

Level Descriptor Comparisons

PIAAC Level	Approximate NRS Level(s)	Approximate CEFR Level(s)
Below Level 1 Can read short familiar words. Very low print literacy.	Beginning Literacy, Low Beginning ABE, ESL Beginning Literacy	Pre-A1 to A1
Level 1 Can read short, basic sentences. Low print literacy.	High Beginning ABE, Low Intermediate ABE, ESL Low Beginning	A1-A2
Level 2 Moderate print literacy. Can infer basic information from text. Can read moderately complex paragraphs.	Intermediate ABE, High Intermediate ESL	A2-B1
Level 3 Can read complex text, compare, and evaluate evidence and make inferences.	Adult Secondary Education (ASE), Advanced ESL	B1-B2
Level 4 Can articulate complex arguments and ideas in both productive areas of proficiency.	Advanced ASE / Postsecondary Readiness	B2-C1

Level 5 College-ready skills. Can evaluate and process complex text. Capable of higher-order reasoning and thought in target language.	Above ASE / College-level literacy	C1-C2

**This table is a point of reference for the reader when considering Pre-level 1/Level 1, NRS level 1, and CEFR A1 level learners' capacities and limitations.*

It is important to note that these methods do not provide a settled conclusion about all flyers geared toward communities of emergent readers all over the United States, nor do these methods provide internationally generalizable results. However, the techniques and procedures used in this research would be simple to replicate for a much larger data sample in further studies.

Analysis and Findings

It should be noted that our biases in this study include: being TESOL teachers (of both low and high levels of proficiency), speaking English as our L1, and coming from a country where English is the primary language (U.S.) and print literacy is extremely high. This may alter our interpretation of the data. In this study, we also assume that the target population has either physical or digital access to these flyers (i.e., having Facebook accounts). We recognize that literacy may not be the only inhibitor to understanding the flyers; other inhibitors may include lack of cultural knowledge of symbols (like business logos or icons) and geographical knowledge.

The linguistic features of the flyers varied widely. The average total number of words on the flyers ranged from only 12 to 420. This naturally resulted in much smaller font size on the flyers with more words, as well as fewer images. Twelve of the thirty flyers analyzed lacked any images, and 5 flyers lacked symbols, such as logos or calendars. We learned from Bruski and

Carsten that realistic, colored, and iconic images aid reader comprehension, especially among low-level learners. Additionally, over half of the flyers had no QR codes or links present. There was a relatively high average syllable per word count showing an average of 1.6 syllables per word and greater in a third of the flyers. The studies above show that a person who scores an A1 level or lower on the CEFR scale can on average capably read and understand about 1.35 to 1.37 syllables per word (or roughly 137 syllables per 100 words). Our research demonstrates that there are many informational flyers that claim to be geared toward communities of English learners and pre-literate individuals that do not follow patterns that would make them more accessible to those same individuals.

Table 2

Raw Data: Quantified Word Count and Graphic Images

	Average syllables per word	total word count	average sentence length (words)	images	symbols (calendar, logos, clip art, etc)	Title font size	font size of important information (time and date)	QR code or link provided	theme/service pr
Flyer 1	1.5	55	11	2	4	40-48 pt	18-22 pt	0	Health Fair Community Cafe
Flyer 2	1.4	48	14	1	4	32-40	16-18	1	table talk
Flyer 3	1.6	135	11	0	4	24-28	10-12	1	Digital Literacy General Commun
Flyer 4	1.8	38	10	1	12	28-36	8-10	1	Service Provider I
Flyer 5	1.7	67	8	3	1	24-30	12-14	0	Rental Assistance Early Childhood
Flyer 6	1.6	96	12	1	11	28-34	8-10	1	Services Free Clothing & C
Flyer 7	1.4	64	7	0	6	32-40	14-18	1	Event
Flyer 8	1.5	86	11	1	4	28-36	18-22	1	Orthodontic Flyer
Flyer 9	1.4	25	4	1	3	34-42	18-24	0	Free Eye Clinic
Flyer 10	1.3	20	3	1	3	28-36	18-24	0	Free Eye Clinic HR1 Bill and imp
Flyer 11	1.7	420	15	3	0	48-52	8	0	families

									world refugee day
Flyer 12	1.3	12	6	0	1	60-70	32-40	0	celebration
Flyer 13	1.4	20	7	1	0	48-56	28-34	1	US citizenship cla
Flyer 14	1.4	60	8	1	0	72-80	36-42	2	body safety class
Flyer 15	1.5	110	9	3	0	64-72	30-26	2	parenting classes
Flyer 16	1.6	250	12	2	2	54-60	28-32	1	immigration and l services
Flyer 17	1.5	180	10	1	2	58-64	30-34	1	refugee youth lead program
Flyer 18	1.3	25	6	0	2	80-90	36-40	1	teddy bear picnic
Flyer 19	1.4	35	7	1	1	60-68	28-32	0	provo city job pos
Flyer 20	1.5	90	9	1	4	64-72	30-36	1	BYU citizenship c
Flyer 21	1.8	56	8	2	3	40-48	18-24	0	Survival/Wilderne Skills
Flyer 22	1.6	42	15	0	2	36-44	18-22	0	Recreation
Flyer 23	1.5	79	2.75	0	6	32-40	20-28	0	Health Fair (Scree for breast cancer)
Flyer 24	1.5	73	12.5	0	3	34-42	18-24	0	Food Drive
Flyer 25	1.7	34	17	1	8	36-44	22-30	0	Mental Wellness I
Flyer 26	1.6	49	12	0	6	30-38	18-24	0	Community Conn Night
Flyer 27	1.4	18	18	0	1	40-50	20-28	0	Food Drive
Flyer 28	1.4	87	17	0	0	26-32	14-18	0	Food Drive
Flyer 29	1.8	23	5	0	1	28-36	22-30	1	Adult English Lar Program
Flyer 30	1.4	32	7	0	30	36-44	20-28	0	Free Neighborhoo Meet

We created an accessibility rating for each flyer based on our linguistic methodology and the print literacy characteristics. The rating system ranges from very high accessibility to very low with a total of seven rating degrees. We rated flyers 9, 10, 12, 18, 27, 14 as having very high accessibility. The features included large font sizes, minimal text in the main body, realistic pictures, and few to no symbolic images. Flyers nine and ten are for separate community eye clinics but have similar highly accessible features. Both have one clear, realistic image that immediately details the purpose of the flyer. One uses glasses graphics to further drive attention

to the purpose. They both use a big and bold font for the main title and clear font and sizing for the event details. Flyer ten uses no extraneous details or symbolic images that could confuse the reader. Flyer nine is rated lower because it uses symbolic images to represent the organization and its funders. According to Bruski, these types of images are harder to understand and could lead the reader's eyes away from the pertinent information. It is noteworthy that most of the public health related flyers were rated among the most accessible. These organizations are mandated under the "Plain Language" Act of 2010 to use this level of accessibility in their communications with the public. Flyers twelve and eighteen are both social community events open to the public. Flyer twelve advertises for "World Refugee Day." It has the linguistic and visual accessibility associated with this rating and is designed for the refugee language learner in mind. The iconic images include the world globe and people representing varying cultural backgrounds, which are easily understood by a language learner. The main title is big and centered and the pertinent event details are clear and easily readable. This flyer is rated lower than the eye clinic flyers because the purpose of the event isn't immediately clear unless the reader has prior experience at this event. However, the linguistic and stylistic features are consistent with this accessibility level. Flyer eighteen is rated lower than twelve because of the animated cultural symbolic image of Winnie the Pooh and the organization's logo. The purpose of the event may be discernible depending on the reader's linguistic familiarity with "picnic." Flyers twenty-seven and fourteen are rated very high for the same linguistic reasons. However, the use of artistic design elements that are more indexical (Bruski, 2012) skew the reader's eye from the center and main point of interest. The "Body Safety Class" may be culturally unfamiliar and could confuse the reader as to its purpose.

We determined three “highly” accessible flyers. Flyer one is a health fair and uses two real photographs of the event. The event details are big and clear. This is rated lower than the food drive and body safety class because it includes more words and some health vernacular. The photos and title of the flyer make this a highly accessible flyer because the reader will easily pick up on its intent. Flyer thirteen is rated as highly accessible because of its simple format, realistic image, and simple linguistic features. However, it is rated below the others because of its symbolic and indexical images, and stylistic features. It may have a realistic photo but it does not naturally represent a citizenship or civics class unless you are familiar with Mount Rushmore and American history. This level of cultural awareness is not easily accessible. Last among the highest accessible flyers, flyer thirty seems to be linguistically simple and accessible. Yet, there is cultural vocabulary in the title that would not be understood even with a direct translation. In addition, the graphic images around the center are not realistic, they are hyper colored, yet mostly iconic. We rated this flyer as having “high” accessibility due to the simplified linguistic features and style in the center of the flyer. This helps to draw the eye to the pertinent information. The very high and highly rated flyers are more linguistically attainable to learners of English at the PIAAC below-level 1, level 1, NRS pre-beginning and low beginning ESL levels, and CEFR Pre-A1, A1-A2 levels.

Table 3

Accessibility of Community Flyers for Emergent Readers

Flyer	Topic	Accessibility Rating	Primary Strengths	Primary Barriers
10	Free Eye Clinic	Very High	Extremely limited text; prominent event information	None identified

9	Free Eye Clinic	Very High	Minimal text; large title; highly visible event information	Minor symbolic images
12	World Refugee Day Celebration	Very High	Extremely limited text; very large title and event information	Few barriers identified
18	Teddy Bear Picnic	Very High	Minimal text; very large title and event information	Few barriers identified
27	Food Drive	Very High	Very limited text; highly visible information	Few barriers identified
14	Body Safety Class	Very High	Large title; highly visible event details; limited reading demand	Potentially unfamiliar topic-specific terms
1	Health Fair	High	Large title; clearly displayed event information; limited text	Some health-related vocabulary
13	U.S. Citizenship Class	High	Minimal text; large font; QR code for additional information	Some unfamiliar civic vocabulary
30	Free Neighborhood Swap Meet	High	Limited text; highly recognizable community activity	Minor visual clutter from numerous symbols
7	Free Clothing and Gear Event	Moderate-High	Concrete vocabulary; large headings; visual layout	Multiple scheduling details and participation rules
15	Parenting Classes	Moderate-High	Large headings; visual supports; information organized into sections	Moderate text density; multiple program details
19	Provo City Job Posting	Moderate-High	Limited text; large font; straightforward purpose	Employment-related vocabulary; application expectations

21	Survival/Wilderness Skills	Moderate-High	Visual supports; clear activity focus	Specialized outdoor vocabulary
24	Food Drive	Moderate-High	Familiar topic; moderate text load; clear organization	Multiple informational details
26	Community Connection Night	Moderate-High	Community-focused language; moderate text load	Several activity descriptions and informational sections
2	Community Café and Table Talk	Moderate	Clear date, time, and location format	Abstract terms such as recognize and utilize
8	Orthodontic Care	Moderate	Large print; clear contact information	Technical terms such as orthodontic and Invisalign
17	Refugee Youth Leadership Program	Moderate	Large title; visual supports; clear program focus	Moderate text load; abstract leadership-related vocabulary
20	BYU Citizenship Clinic	Moderate	Large event information; QR code support	Civic/legal terminology and multiple service details
22	Recreation Program	Moderate	Low word count; organized information	Long sentence structure may increase processing demands
23	Health Fair (Breast Cancer Screenings)	Moderate	Clearly identified event; health-related visuals	Medical terminology and longer sentences
28	Food Drive	Moderate	Familiar topic; manageable word count	Small font and relatively long sentences
29	Adult English Language	Moderate	Short text; QR code support	Educational terminology and higher syllable

	Program			complexity
4	Community Programs and Services	Moderate-Low	Large headings and visual organization	Multiple unfamiliar service-related terms
25	Mental Wellness Fair	Moderate-Low	Large headings; numerous visual cues	Mental health terminology; relatively complex language
5	Rental Assistance	Low	Bulleted information	Specialized vocabulary (eligibility, mediation, expungement)
6	Early Childhood Services	Low	Clear website and visual categories	Abstract and multisyllabic vocabulary
16	Immigration and Legal Services	Low	Large font and visual organization; contact information provided	High text density; legal and immigration terminology
3	Digital Literacy	Very Low	Organized by day and time	High text density; multiple locations; large information load
11	HR1 Bill and Impact on Families	Very Low	Large title; some visual supports	Extremely high text density; policy-related vocabulary; lengthy sentences

Flyers numbered 7, 15, 19, 21, 24, and 26 were categorized as having moderate-high accessibility to English learners. These often demonstrated good accessibility in many facets, only lacking in one area, often clear vocabulary or learner-appropriate density. Flyers focused on parenting classes, job postings, and survival skills shared some common strengths, featuring clear organization, large text, and visuals, while also including specialized vocabulary that may be difficult for an inexperienced English learner to understand. Interestingly, the other three

flyers marked with moderate-high accessibility had opposite strengths and weaknesses. Flyers advertising community connection, food drives, and free clothing had clear vocabulary in common, while lacking transparent formatting, as well as including some wordy extra details.

We ranked eight flyers as ‘Moderate’ (without ‘high’ or ‘low’). These flyers demonstrated some common traits, including longer sentences, and some crowding of symbols and stylistically distracting visuals, balanced by some easily accessible traits such as good organization, shorter word count, and large font size. For example, the flyer for the adult English language program was marked as moderate, as it featured a QR code and visual supports, but was complex and sentences often included a high number of multi-syllabic words. A food drive flyer was marked as moderate due to a simple topic and minimal text in contrast to small font size and relatively long sentences. Another flyer for a community recreation program exhibited good organization, but long sentences that may lead to processing difficulties for less experienced speakers of English. Yet another health-related flyer was identified as moderately accessible because it used too much specific language–medical terminology–and longer sentences. Flyers for a BYU citizenship clinic and another for orthodontic care both demonstrated large font size and clear organization, but contained an excess amount of civic and technical terms respectively. Finally, two more flyers for community events and refugee leadership programs showed clear formatting and visual aids, while lacking clarity in vocabulary.

Although the moderately rated flyers’ intended audience were not always language learners in the community, many of their services target the lower-income population, of which, as noted in the introduction, a significant portion are English language learners. These organizations would provide a greater public benefit if they modified the formatting of their flyers to include far fewer words, a bold specific message, less stylistic distraction, and a clear

way to contact for more information. This advice applies especially to the organizations whose flyers were rated as low and very low. The moderately high and moderate rated flyers are more linguistically attainable to learners of English at the PIAAC levels 1-2, NRS low beginning-high beginning ESL levels, and CEFR A1-B1 levels.

We ranked seven flyers as low accessibility (including degrees Moderate-Low, Low, and Very Low). These demonstrated some similar accessibility barriers: dense, long text and advanced terminology above the reading English level of their target audience. For example, the immigration and legal services flyer had terminology like ‘consular processing’ and ‘asylum’ that would even be hard for a higher English level reader to understand.

Consistent with the findings discussed in the literature review, Table 3 suggests that flyers varied in their accessibility and for different reasons for English language learners. This shows that at least half of the collected flyers may be categorized as linguistically and visually accessible to low-level English learners in the greater Provo/Orem area. These flyers contained too much specialized vernacular, lengthy sentences and paragraphs, and lack of focus in the message. These services are highly relevant to language learners. More guidance for human-service organizations is needed to provide a greater benefit to the whole community, including and most especially, the English language learning community. The moderately low, low, and very low rated flyers are more linguistically attainable to learners of English at the PIAAC levels 2-3, NRS high beginning-intermediate ESL levels, and CEFR B1-C1 levels.

Discussion and Reflection

These findings illuminate both strengths and limitations to this study. In general, some of the flyers could be considered accessible to lower-literacy students. This suggests that

community organizations (included in the collected samples) tended to use simple, understandable language targeted for their intended audiences. For adult English learners, this finding may mean that learners in lower PIAAC levels and CEFR reading abilities may have more access to community events because they may understand the information on the flyers. Moreover, this study not only examined textual accessibility, but also visuals. We considered word count, syllabic complexity, sentence length, font size of information of the flyers, and literacy level. This provides a more comprehensive analysis of accessibility because interpreting a flyer as an English language learner depends on both language and how information is visually presented.

Despite its strengths, this study encountered limitations. Thirty flyers is a relatively low sample size. Therefore, these findings cannot be generalized to all community flyers, all cities, all services, nor all communication contexts. This may limit external validity. Also, these flyers were not collected randomly; they were collected through convenience sampling, meaning that the data set may have overrepresentation. Organizations that actively distribute flyers or post them on social media mainly constituted the samples. Additionally, difficult vocabulary may have increased the PIAAC levels of the flyers. Comprehension became harder due to the discipline-vernacular (i.e., health/medical, legal, immigration, and public policy terminology). Though the sentence syntax was simple, the vocabulary made it more suited for a higher literacy level. Furthermore, the PIAAC level classifications were not administered to actual students nor was official PIAAC assessment used. Instead, the level categorization was assigned based on characteristics of the text, such as information density, sentence complexity, and vocabulary. Consequently, these level categories for the flyers should be viewed as estimates rather than official PIAAC scores.

Considering this discussion, companies should take care and be aware of difficulties that may arise when creating flyers for new speakers of English. If organizations want to reach their target audience, vocabulary should be simplified to match their understanding and less dense text should be on the flyers. Flyers should especially be accessible if there is pertinent information to the audience (i.e., legal, medical, taxes, etc.). Other revisions should include simple language and clear images. Such modifications could increase access to important community resources for English language learners.

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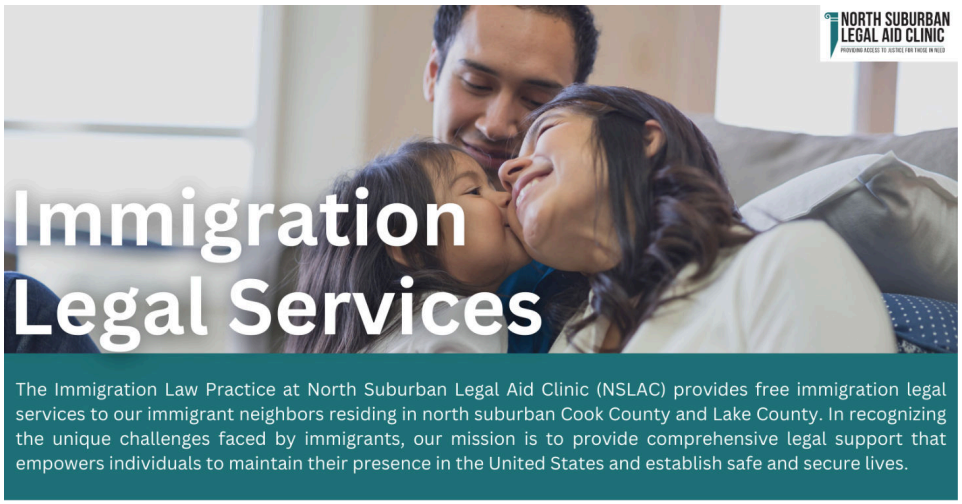
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Appendix

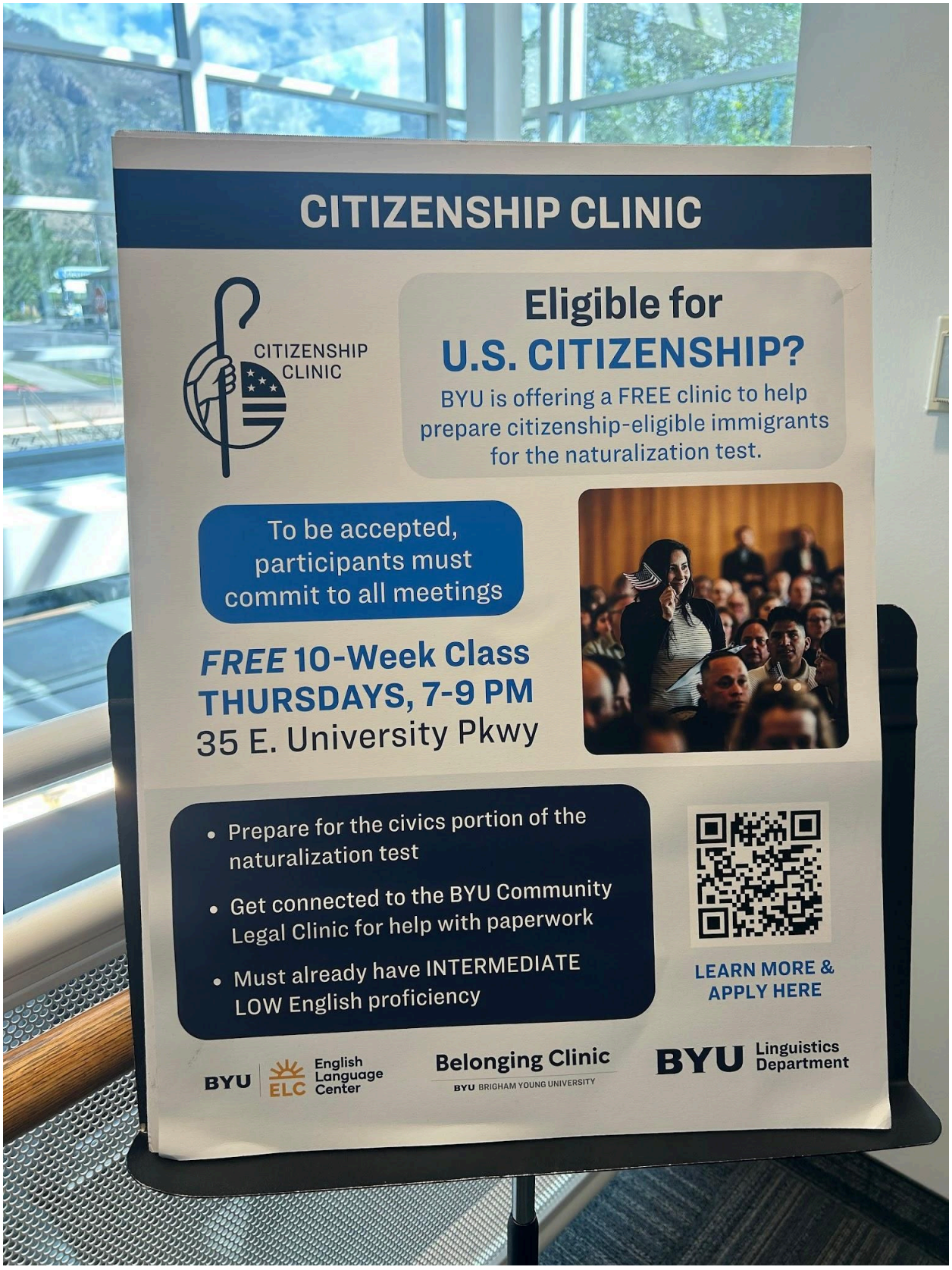
Flyers

Thirty collected flyers for the current research, "Accessibility of Community Flyers: A Content Analysis of Social Service Flyers in Utah County."

Low Accessibility Flyer



Moderate accessibility flyer



High accessibility flyer

